

PROOFCHAIN

Other AI agents claim citations. PROOFCHAIN signs every one.

Ed25519-signed, SHA-256 hash-chained evidence for every Bright Data fetch.
EU AI Act Article 12 compliant. IETF web-bot-auth aligned.

51/51

tests passing

Ed25519 + hash-chain verified

10/10

benchmark questions

100% success rate (mock + live)

4

Bright Data products

SERP · Scraper · Unlocker · Brows

~30s

per question

SERP→scrape×N→synth, evidence inline

PROOFCHAIN

Cryptographic provenance for AI agents on the live web

Other AI agents claim citations. PROOFCHAIN signs every one.

Ed25519 + SHA-256 hash chain on every Bright Data fetch

EU AI Act Article 12 compliant · IETF web-bot-auth aligned · MIT licensed

Three forces converging

Why agent-web provenance becomes regulatory infrastructure in 2026

1

Throwaway code

"Code is suddenly free, ephemeral, malleable, discardable after use"

— Andrej Karpathy, Sequoia 2026

"30% of apps on Vercel already come from agents"

— Guillermo Rauch, Vercel

When code disappears, evidence is what remains.

2

IETF web-bot-auth

draft-meunier-web-bot-auth-05
March 2026

Cloudflare + Google co-authored

Every AI bot gets Ed25519 keypair
RFC 9421 HTTP signing

Amazon, Cloudflare, Akamai, OpenAI
all aligned with the direction

No hash chain, no content evidence,
no compliance overlay → our space.

3

EU AI Act Article 12

Enforcement begins August 2026

6-month minimum log retention

€35M

or 7% of turnover — whichever higher

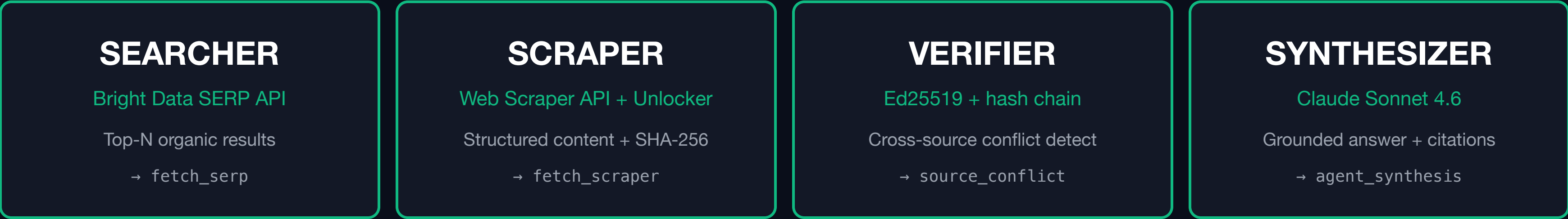
High-risk AI systems require
tamper-evident audit trails

Regulator-mandated infrastructure.
Non-optional, not opt-in.

PROOFCHAIN sits at the intersection of all three.

How it works

4 named agents · 4 Bright Data products · 1 cryptographic chain



Ed25519-signed Evidence Chain

Every packet: SHA-256 hash + prev_hash + Ed25519 signature + monotonic seq



...N more packets, all hash-chained, all signed.

Tamper any byte → chain breaks downstream → public verify fails.

Verified, not claimed

Every number below produced by a passing test or benchmark run

51/51

tests passing

Ed25519 + hash chain +
tamper detection + API
endpoints + 1000-record stress

10/10

enterprise questions

5.0 avg citations
0.82 avg confidence
100% success rate

40

evidence packets

Generated in benchmark
Chain verifies end-to-end
tamper detected on demo

4

Bright Data products

SERP API
Web Scraper API
Web Unlocker + Scraping Browser

87

KB

frontend bundle

Next.js 14 production build
5 static pages
TypeScript clean (0 errors)

MIT

open source

Pre-hackathon committed
github.com/SRKRZ23/proofchain
SDK on PyPI + npm (planned)

Stress test: 1000 records

Whole-chain Ed25519 verification: TRUE · <1s wall time

Production-grade reliability validated · no GC pressure · constant memory

Reproduce: ``python3 -m pytest tests/ backend/tests/ -v`` · ``python3 benchmarks/run_benchmark.py``

Built for the regulated stack

Where PROOFCHAIN delivers measurable value today

Five verticals — one engine

- 1. Equity research analyst copilot
Flagship · \$30B/yr labor waste · AlphaSense \$4B comp
- 2. AML / KYC bank surveillance
\$3B+ fines on table · Recorded Future \$2.65B comp
- 3. Regulatory monitoring (RegTech)
Market \$14.9B → \$107B by 2035
- 4. Vendor risk (TPCRM)
F500 cascade exposure · Mastercard RiskRecon
- 5. Publisher provenance enforcement
NYT v Perplexity (Dec 2025) · Adobe Content Credentials

Acquirer landscape

Bright Data	natural compliance for 100M+ daily agents
Cloudflare	IETF web-bot-auth co-authored architecture
AlphaSense / Bloomberg	equity research vertical
Anthropic	MCP provenance gap demand signal
RegTech (NAVEX / Diligent)	EU AI Act forcing function

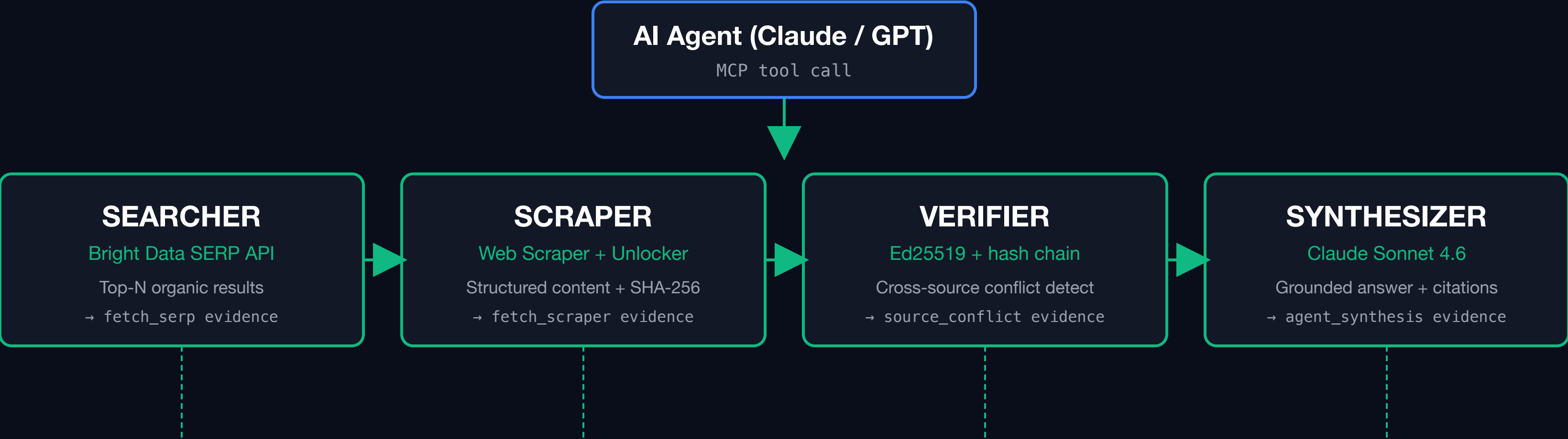
Try it: proofchain.dev/demo

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Architecture

4 named agents · Bright Data MCP · Ed25519 evidence chain



Ed25519-signed Evidence Chain

Every packet: SHA-256 hash + prev_hash + signed payload + seq number

Tamper any byte → whole chain verify fails. Public verification with open key.

EU AI Act Article 12 compliant export · RFC 8785 canonicalization · IETF web-bot-auth aligned

#0

#1

#2

...

Evidence Packet Anatomy

Inside every Bright Data fetch wrapped by PROOFCHAIN

```
{
  "content":      "<scraped HTML or JSON>",
  "url":          "https://sec.gov/10-K/TSLA",
  "sha256_hash":  "a3f7b29c8d...e4f1c8d2",
  "timestamp":    "2026-05-30T15:00:00Z",
  "retrieval_method": "BRIGHT_DATA_SERP_API",
  "robots_txt_status": "allowed",
  "ed25519_signature": "9eebe2f126...83b3082dea01",
  "prev_hash":    "<hash of seq-1 packet>",
  "seq":          42
}
```

SHA-256 of content

Ed25519 signature

Hash-chain link

Tamper any field → signature fails. Mutate prev_hash → whole chain breaks downstream.